1.Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

GPU Identification Report

GPU Brand: Intel

GPU Model Number: Intel® UHD Graphics

Chip Type: Intel® Raptor Lake-S Mobile Graphics Controller

GPU Type: Integrated GPU

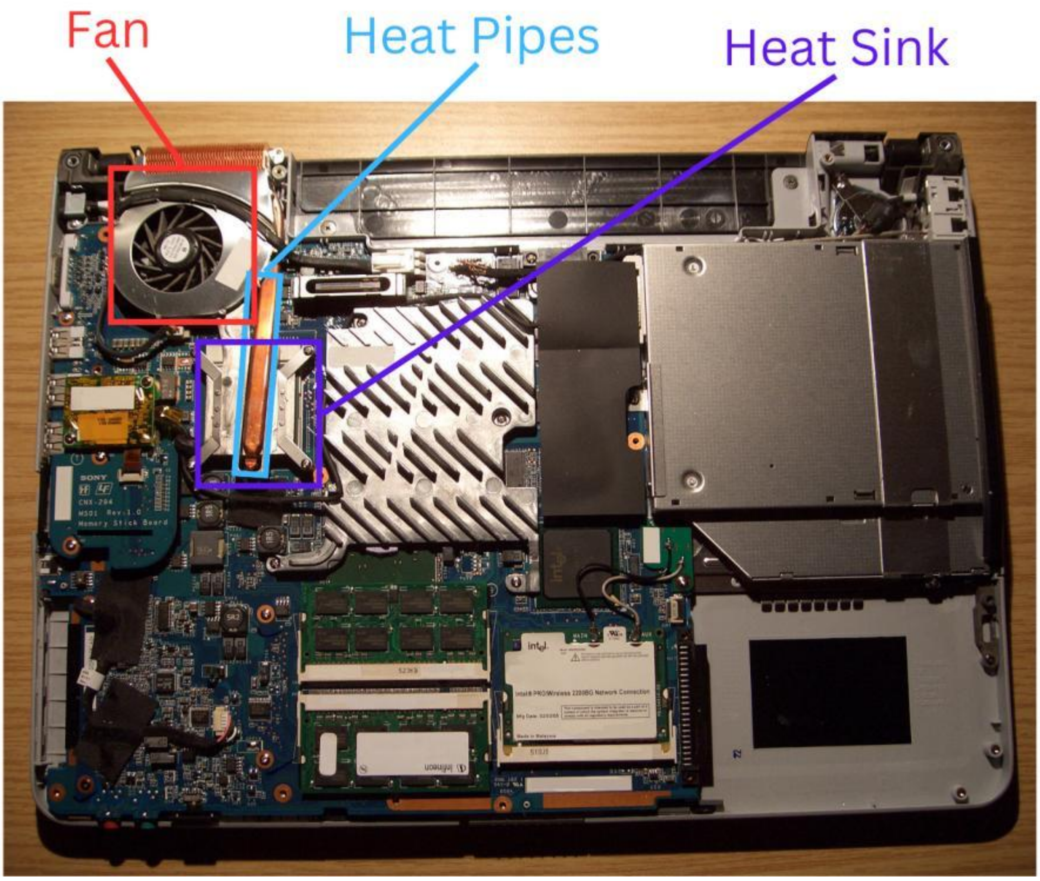
Description

The computer is equipped with **Intel® UHD Graphics**, an **integrated graphics processing unit (GPU)** manufactured by Intel Corporation. This GPU is built into the processor and shares system memory (RAM) with the computer instead of having its own dedicated video memory. It is suitable for everyday tasks such as web browsing, office applications, multimedia playback, and light gaming.

System Information (from DirectX Diagnostic Tool)

- **Name:** Intel® UHD Graphics
- **Manufacturer:** Intel Corporation
- **Chip Type:** Intel® Raptor Lake-S Mobile Graphics Controller
- **Display Memory:** 128 MB
- **Approximate Total Graphics Memory:** 8162 MB
- **Direct3D Acceleration:** Enabled

2.Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



Labels

1. **CPU Fan** – Cools the processor by pushing hot air out.
2. **Heatsink** – Absorbs heat from the CPU.
3. **Heat Pipe** – Transfers heat from CPU to heatsink.
4. **Air Vent** – Releases hot air outside the laptop.

3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	**Air Cooling**	**Liquid Cooling**
Cost	Less expensive and budget-friendly.	More expensive due to additional components.
Cooling Efficiency	Good for normal gaming and everyday use.	Better cooling performance, especially under heavy workloads.
Maintenance	Requires very little maintenance, mainly dust cleaning.	Requires regular maintenance and checking for leaks.
Installation	Easy to install and replace.	More complex installation process.
Noise Level	Can be noisy at high fan speeds.	Usually quieter because fans run at lower speeds.
Size	Takes less space inside the PC.	Requires extra space for radiator, tubing, and pump.

4.Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

Choice: Liquid Cooling for GPU

I would choose **Liquid Cooling** for my gaming PC because I plan to stream IPL matches and play graphics-heavy games at the same time. Liquid cooling provides better cooling efficiency and keeps the GPU temperature lower during long gaming sessions. It also operates more quietly than air cooling, which is beneficial while streaming. Although it is more expensive, the improved performance and temperature control make it the better choice for a high-performance gaming PC.